

Motivation for feature-based alignment: Image mosaics

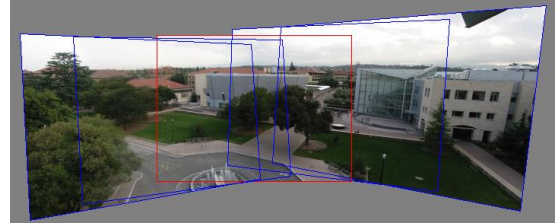


Image from http://graphics.cs.uni.edu/courses/CS-463/2010_fall/

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Feature-based alignment



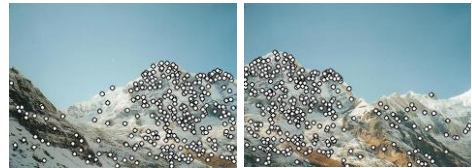
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Source: L. Leifert

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Feature-based alignment



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Source: L. Leifert

- Extract features

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Feature-based alignment



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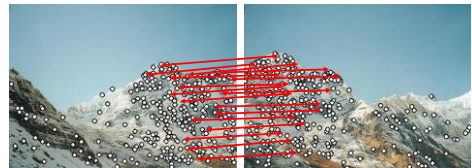
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- Extract features
- Compute matches

Feature-based alignment



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Source: L. Leifert

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- Extract features
- Compute matches
- Filter matches

Feature-based alignment

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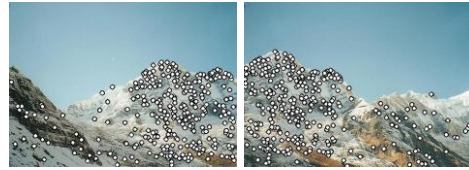


- Extract features
- Compute matches
- Filter matches
- Compute transformation

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Feature-based alignment

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- Extract features

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What points would you choose?

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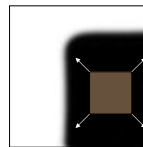


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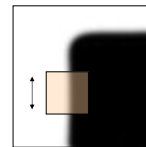
Distinctive interest points

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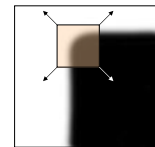
- We should easily recognize the point by looking through a small window
- Shifting a window in *any* direction should give a *large* change in intensity



"flat" region:
no change in
all directions



"edge":
no change
along the edge
direction



"corner":
significant
change in all
directions

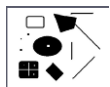
Slide credit: Alexey Efros, Darya Prokopy, David Sussman 11/12/2024 Jürgen Gall – Institute of Computer Science III – Computer Vision Group

Corners as distinctive interest points

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$$M = \sum w(x, y) \begin{bmatrix} I_x I_x & I_x I_y \\ I_x I_y & I_y I_y \end{bmatrix}$$

2 x 2 matrix of image derivatives (averaged in neighborhood of a point).



Notation:

$$I_x \Leftrightarrow \frac{\partial I}{\partial x}$$

$$I_y \Leftrightarrow \frac{\partial I}{\partial y}$$

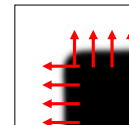
$$I_x I_y \Leftrightarrow \frac{\partial I}{\partial x} \frac{\partial I}{\partial y}$$

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What does this matrix reveal?

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First, consider an axis-aligned corner:



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What does this matrix reveal?

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First, consider an axis-aligned corner:

$$M = \sum \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

This means dominant gradient directions align with x or y axis

Look for locations where **both** λ 's are large.

If either λ is close to 0, then this is **not** corner-like.

What if we have a corner that is not aligned with the image axes?

Klaus Graumann

What does this matrix reveal?

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Since M is symmetric, we have $M = X \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} X^T$



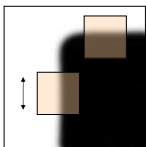
$$Mx_i = \lambda_i x_i$$

The *eigenvalues* of M reveal the amount of intensity change in the two principal orthogonal gradient directions in the window.

Klaus Graumann

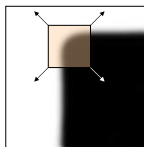
Corner response function

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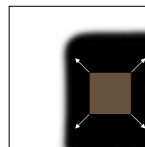
"edge":

$$\lambda_1 \gg \lambda_2 \\ \lambda_2 \gg \lambda_1$$



"corner":

$$\lambda_1 \text{ and } \lambda_2 \text{ are large,} \\ \lambda_1 \sim \lambda_2;$$



"flat" region

$$\lambda_1 \text{ and } \lambda_2 \text{ are} \\ \text{small;}$$

$$f = |M| - k (\text{Tr}(M))^2 = \lambda_1 \lambda_2 - k (\lambda_1 + \lambda_2)^2$$

$$k \approx 0.04$$

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Harris corner detector

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- 1) Compute M matrix for each image window to get their *cornerness* scores.
- 2) Find points whose surrounding window gave large corner response ($f > \text{threshold}$)
- 3) Take the points of local maxima, i.e., perform non-maximum suppression

Klaus Graumann

Harris Detector

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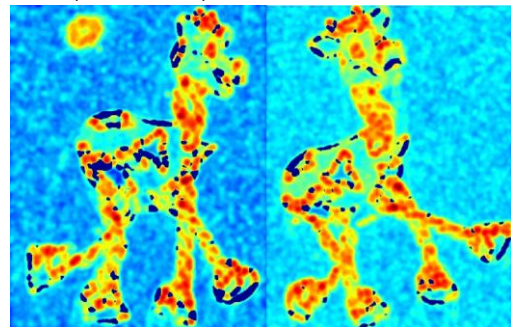


Klaus Graumann

Harris Detector

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Compute corner response f



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Harris Detector

Find points with large corner response: $f > \text{threshold}$



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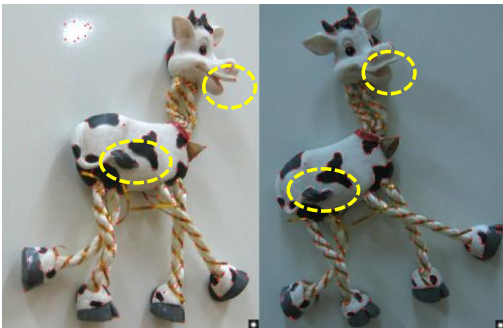
Harris Detector

Take only the points of local maxima of f



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Harris Detector



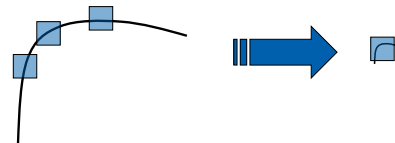
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Properties of the Harris corner detector

- Rotation invariant? Yes

$$M = X \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} X^T$$

- Scale invariant? No



All points will be classified as edges

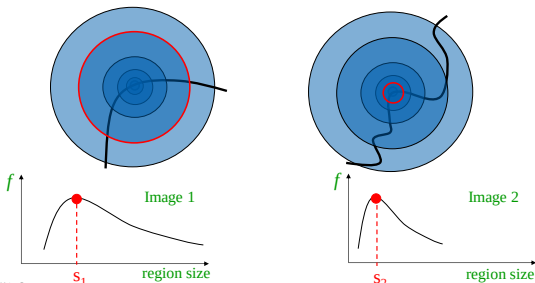
Corner !

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Automatic scale selection

Intuition:

- Find scale that gives local maxima of some function f in both position and scale.



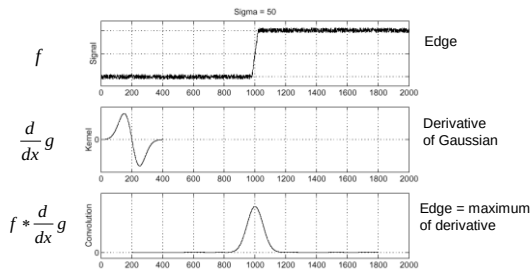
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Scale invariant blob detection



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Recall: Edge detection

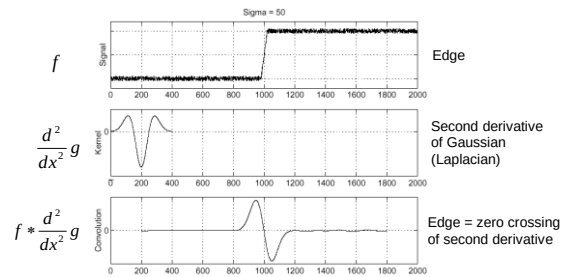


Source: S. Seitz

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Recall: Edge detection



Source: S. Seitz

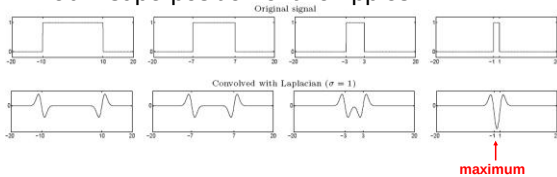
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From edges to blobs



- Edge = ripple
- Blob = superposition of two ripples



Spatial selection: the magnitude of the Laplacian response will achieve a maximum at the center of the blob, provided the scale of the Laplacian is “matched” to the scale of the blob

Lars Lovell

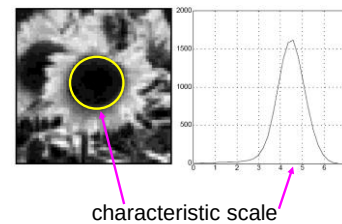
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Blob detection in 2D



We define the *characteristic scale* as the scale that produces peak of Laplacian response



Lars Lovell

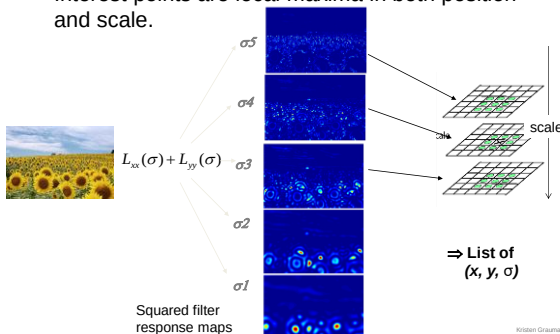
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Scale invariant interest points



Interest points are local maxima in both position and scale.

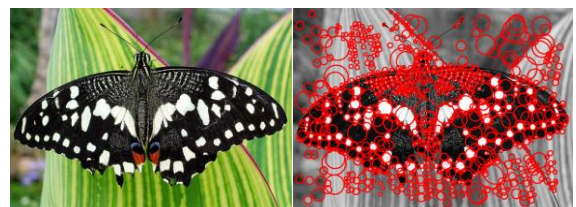


Wojciech Gajda

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Scale-space blob detector



Lars Lovell

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Feature-based alignment

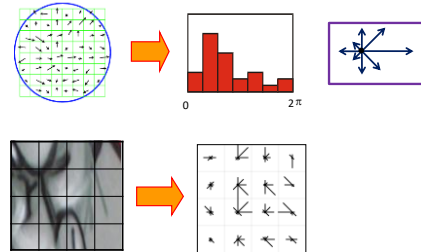


- Extract features
- Compute matches

Source: L. Siskind

SIFT descriptor [Lowe 2004]

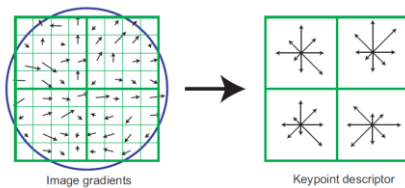
- Use histograms to bin pixels within sub-patches according to their orientation.



Source: Grauman

SIFT descriptor [Lowe 2004]

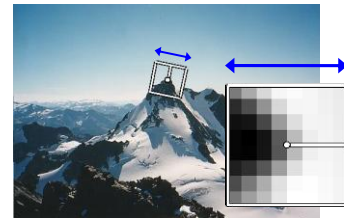
- 128 D vector (4x4x8)



- Normalize to 1 (robust to illumination changes)
- Entries above 0.2 are set to 0.2 (reduce impact of very strong gradients e.g. shadow)
- Normalize to 1

Source: Grauman

Making descriptor rotation invariant



- Rotate patch according to its dominant gradient orientation
- This puts the patches into a canonical orientation.

Source: Grauman

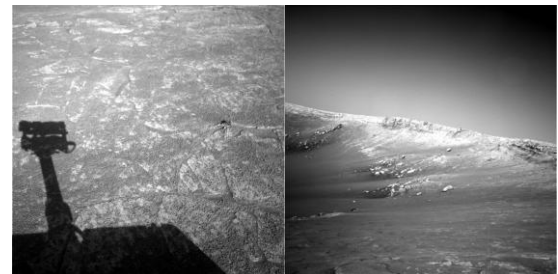
SIFT descriptor [Lowe 2004]

- Extraordinarily robust matching technique
 - Can handle changes in viewpoint
 - Up to about 60 degree out of plane rotation
 - Can handle significant changes in illumination
 - Sometimes even day vs. night (below)
 - Fast and efficient



Source: Grauman

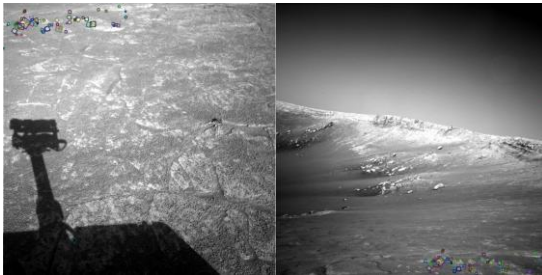
Example



NASA Mars Rover images

Source: Grauman

Example



NASA Mars Rover images
with SIFT feature matches

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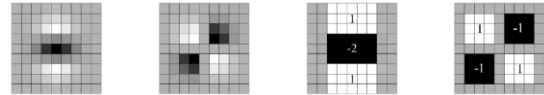
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Other feature descriptors

SURF: Approximate second order derivatives of Gaussian



Fast to compute with integral images

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Feature descriptors

- Countless number of SIFT variations
- SIFT sometimes too slow, many approximations for real-time applications
- Color versions exist

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Feature-based alignment



- Extract features
- Compute matches

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Source: L. Leibe

19

Matching local features



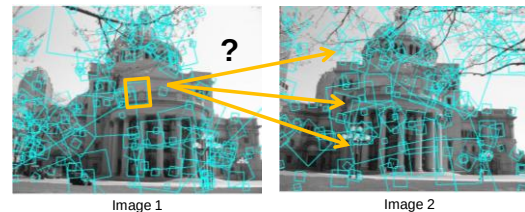
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Matching local features



To generate **candidate matches**, find patches that have the most similar descriptor (e.g., Euclidean distance)
Simplest approach: compare them all, take the closest (or closest k , or within a thresholded distance)

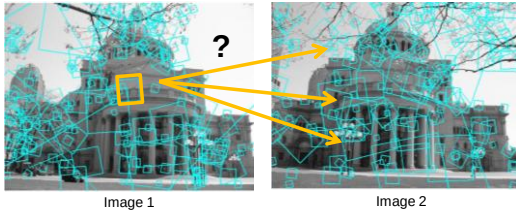
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Matching local features

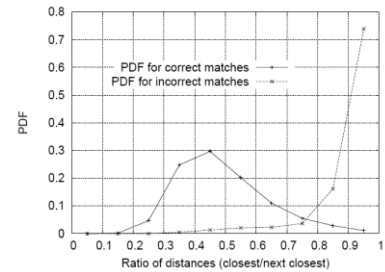


To add robustness to matching, can consider **ratio** :
 distance to best match / distance to second best match
 If low, first match looks good.
 If high and close to one, could be ambiguous match.

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Matching SIFT Descriptors

- Nearest neighbor (Euclidean distance)
- Threshold ratio of nearest to 2nd nearest descriptor



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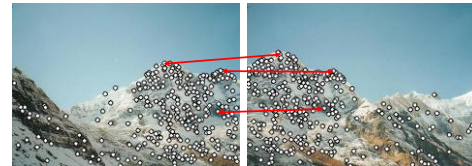
Robust feature-based alignment



- Extract features
- Compute *putative matches*

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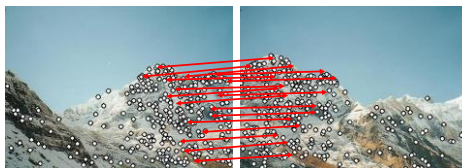
Robust feature-based alignment



- Extract features
- Compute *putative matches*
- Loop:
 - Hypothesize transformation T (small group of putative matches that are related by T)

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Robust feature-based alignment



- Extract features
- Compute *putative matches*
- Loop:
 - Hypothesize transformation T (small group of putative matches that are related by T)
 - Verify transformation (search for other matches consistent with T)

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Robust feature-based alignment

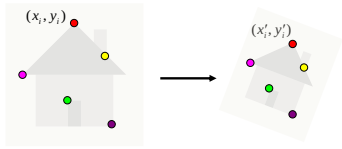


- Extract features
- Compute *putative matches*
- Loop:
 - Hypothesize transformation T (small group of putative matches that are related by T)
 - Verify transformation (search for other matches consistent with T)

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Recall: Fitting an affine transformation





$$\begin{bmatrix} x'_i \\ y'_i \end{bmatrix} = \begin{bmatrix} m_1 & m_2 \\ m_3 & m_4 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} + \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$\begin{bmatrix} x_i & y_i & 0 & 0 & 1 & 0 \\ 0 & 0 & x_i & y_i & 0 & 1 \end{bmatrix} \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ m_4 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} x'_i \\ y'_i \end{bmatrix}$$

Source: K. Grauman

Recall: Least squares problem

A $m \times n$ matrix of rank n :

$$Ax = b$$

Solution with pseudo-inverse:

$$A^T(Ax - b) = 0$$

$$x = (A^T A)^{-1} A^T b$$

Pseudo-inverse A^+

Solution with SVD:

- SVD: $A = UDV^T$; $A^+ = VD^{-1}U^T$
 - $b' = U^T b$
 - $y_i = b'_i / D_{ii}$
 - $x = Vy$
- There are different ways to solve the problem. The optimal method depends on size and sparseness of A
- Specific C++ linear algebra libraries exist!

Outliers



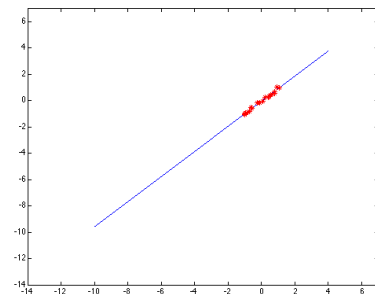
Outliers can hurt the quality of our parameter estimates, e.g.,

- an erroneous pair of matching points from two images
- an edge point that is noise, or doesn't belong to the line we are fitting.



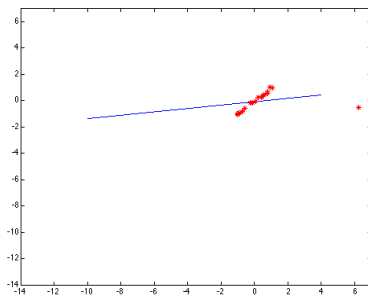
Wilhelm Grunert

Outliers affect least squares fit



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Outliers affect least squares fit



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Random Sample Consensus (RANSAC)

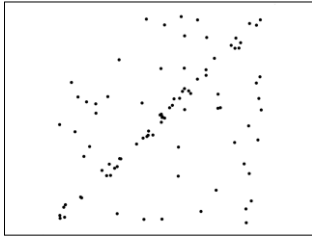
RANSAC loop:

1. Randomly select a *seed group* of points on which to base transformation estimate (e.g., a group of matches)
2. Compute transformation from seed group
3. Find *inliers* to this transformation
4. If the number of inliers is sufficiently large, re-compute estimate of transformation on all of the inliers

Keep the transformation with the largest number of inliers

Wilhelm Grunert

RANSAC for line fitting example



Source: R. Szeliski

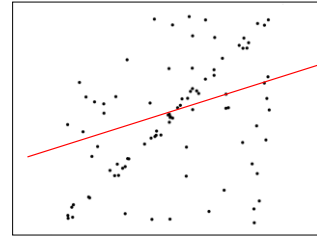
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12

RANSAC for line fitting example



Least-squares fit

Source: R. Szeliski

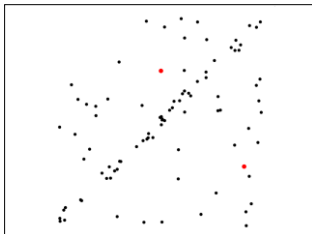
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RANSAC for line fitting example



1. Randomly select minimal subset of points

Source: R. Szeliski

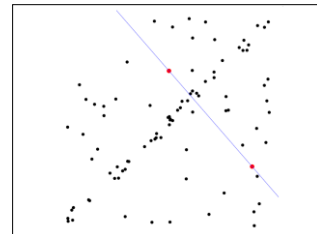
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14

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model

Source: R. Szeliski

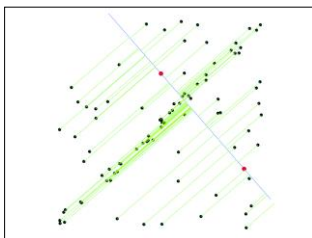
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15

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function

Source: R. Szeliski

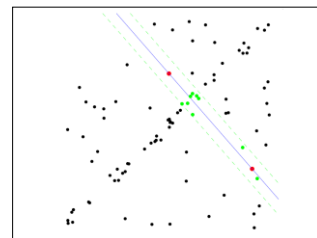
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16

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function
4. Select points consistent with model

Source: R. Szeliski

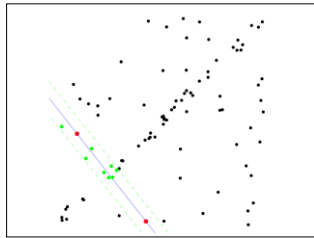
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17

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function
4. Select points consistent with model
5. Repeat *hypothesize-and-verify* loop

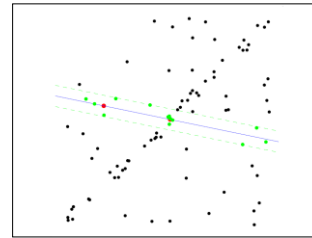
Source: R. Szeliski

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81

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function
4. Select points consistent with model
5. Repeat *hypothesize-and-verify* loop

Source: R. Szeliski

Lars Leibe

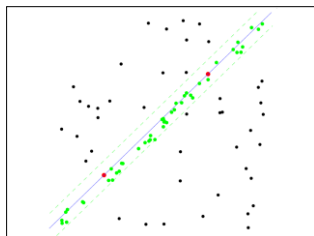
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82

RANSAC for line fitting example



Uncontaminated sample



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function
4. Select points consistent with model
5. Repeat *hypothesize-and-verify* loop

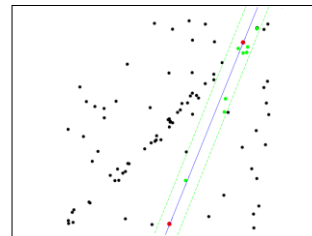
Source: R. Szeliski

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83

RANSAC for line fitting example



1. Randomly select minimal subset of points
2. Hypothesize a model
3. Compute error function
4. Select points consistent with model
5. Repeat *hypothesize-and-verify* loop

Source: R. Szeliski

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RANSAC for line fitting

Repeat N times:

- Draw s points uniformly at random
- Fit line to these s points
- Find inliers to this line among the remaining points (i.e., points whose distance from the line is less than t)
- If there are d or more inliers, accept the line and refit using all inliers

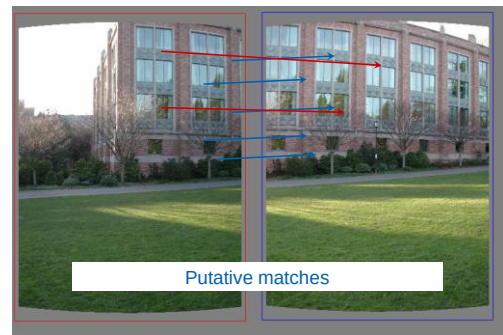
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RANSAC example: Translation



Putative matches

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Source: Marc Stalder

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RANSAC example: Translation

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RANSAC example: Translation

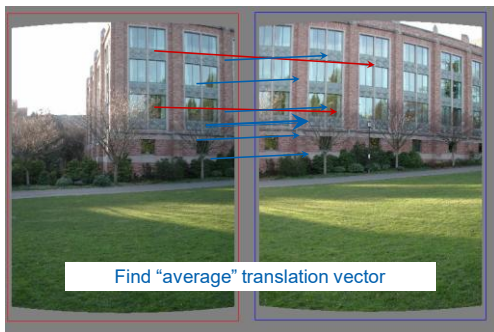
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RANSAC example: Translation

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How Many Samples?

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On average

N ... number of point
 l ... number of inliers
 m ... size of the sample

$$P(\text{good}) = \frac{\binom{l}{m}}{\binom{N}{m}} = \prod_{j=0}^{m-1} \frac{l-j}{N-j}$$

mean time before the success
 $E(k) = 1 / P(\text{good})$

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How Many Samples?

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With confidence p How large k ?

... to hit at least one pair of points on the line l with probability larger than p (0.95)

Equivalently

... the probability of not hitting any pair of points on l is $\leq 1 - p$

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How Many Samples?

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With confidence p

N ... number of point
 l ... number of inliers
 m ... size of the sample

$$P(\text{good}) = \frac{\binom{l}{m}}{\binom{N}{m}} = \prod_{j=0}^{m-1} \frac{l-j}{N-j}$$

 $P(\text{bad}) = 1 - P(\text{good})$

$$P(\text{bad } k \text{ times}) = (1 - P(\text{good}))^k$$

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How Many Samples?



With confidence p

$$P(\text{bad } k \text{ times}) = (1 - P(\text{good}))^K \leq 1 - p$$

$$k \log(1 - P(\text{good})) \leq \log(1 - p)$$

$$k \geq \log(1 - p) / \log(1 - P(\text{good}))$$

$$P(\text{good}) = \frac{\binom{I}{m}}{\binom{N}{m}} = \prod_{j=0}^{m-1} \frac{I - j}{N - j}$$

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72

How Many Samples


 $I/N [\%]$

	15%	20%	30%	40%	50%	70%
2	132	73	32	17	10	4
4	5916	1871	368	116	46	11
7	$1.75 \cdot 10^6$	$2.34 \cdot 10^5$	$1.37 \cdot 10^4$	1827	382	35
8	$1.17 \cdot 10^7$	$1.17 \cdot 10^6$	$4.57 \cdot 10^4$	4570	765	50
12	$2.31 \cdot 10^{10}$	$7.31 \cdot 10^8$	$5.64 \cdot 10^6$	$1.79 \cdot 10^5$	$1.23 \cdot 10^4$	215
18	$2.08 \cdot 10^{15}$	$1.14 \cdot 10^{13}$	$7.73 \cdot 10^9$	$4.36 \cdot 10^7$	$7.85 \cdot 10^5$	1838
30	∞	∞	$1.35 \cdot 10^{16}$	$2.60 \cdot 10^{12}$	$3.22 \cdot 10^9$	$1.33 \cdot 10^8$
40	∞	∞	∞	$2.70 \cdot 10^{16}$	$3.29 \cdot 10^{12}$	$4.71 \cdot 10^9$

Size of the sample m

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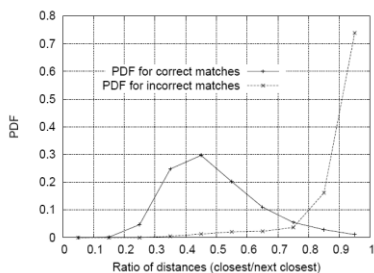
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74

Recall: Matching SIFT Descriptors



- Nearest neighbor (Euclidean distance)
- Threshold ratio of nearest to 2nd nearest descriptor



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How Many Samples


 $I/N [\%]$

	15%	20%	30%	40%	50%	70%
2	132	73	32	17	10	4
4	5916	1871	368	116	46	11
7	$1.75 \cdot 10^6$	$2.34 \cdot 10^5$	$1.37 \cdot 10^4$	1827	382	35
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Size of the sample m

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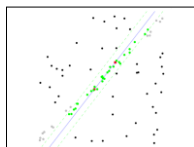
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RANSAC pros and cons



- Pros
 - Simple and general
 - Applicable to many different problems
 - Often works well in practice
- Cons
 - Doesn't work well for low inlier ratios (too many iterations, or can fail completely)
 - Can't always get a good initialization of the model based on the minimum number of samples



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Sergey Lashin

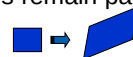
72

Recall: 2D Affine Transformations



$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

- Affine transformations are combinations of
 - Linear transformations, and
 - Translations
- Parallel lines remain parallel



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Source: K. Grauman

74

Projective Transformations

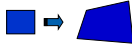


Projective transformations:

- Affine transformations, and
- Projective warps

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Parallel lines do not necessarily remain parallel



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Mosaics



Obtain a wider angle view by combining multiple images.

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How to stitch together a panorama (a.k.a. mosaic)?



Basic Procedure

- Take a sequence of images from the same position
 - Rotate the camera about its optical center
- Compute transformation between second image and first
- Transform the second image to overlap with the first
- Blend the two together to create a mosaic
- (If there are more images, repeat)

...but **wait**, why should this work at all?

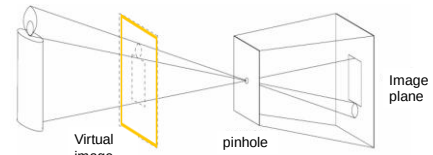
- What about the 3D geometry of the scene?
- Why aren't we using it?

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Pinhole camera



Pinhole camera is a simple model to approximate imaging process, perspective **projection**.



If we treat pinhole as a point, only one ray from any given point can enter the camera.

Fig from Forgy and Ponce 11/12/2024 Jürgen Gall – Institute of Computer Science III – Computer Vision Group 82

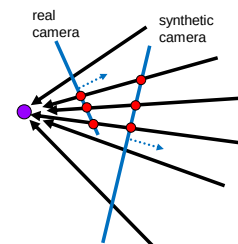
Mosaics



Obtain a wider angle view by combining multiple images.

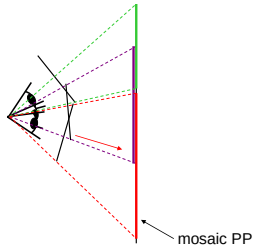
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Mosaics: generating synthetic views



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Image reprojection



The mosaic has a natural interpretation in 3D

- The images are reprojected onto a common plane
- The mosaic is formed on this plane
- Mosaic is a *synthetic wide-angle camera*

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84

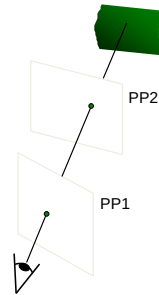
Image reprojection



- How to relate two images from the same camera center? How to map a pixel from PP1 to PP2?

- Cast a ray through each pixel in PP1
- Draw the pixel where that ray intersects PP2

- Rather than thinking of this as a 3D reprojection, think of it as a 2D image warp from one image to another.



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Source: Aljoscha Ethos

85

Image reprojection: Homography

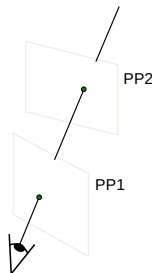


A projective transform is a mapping between any two PPs with the same center of projection

Called **Homography**

$$\begin{bmatrix} wx' \\ wy' \\ w' \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\mathbf{p}' = \mathbf{H} \mathbf{p}$$



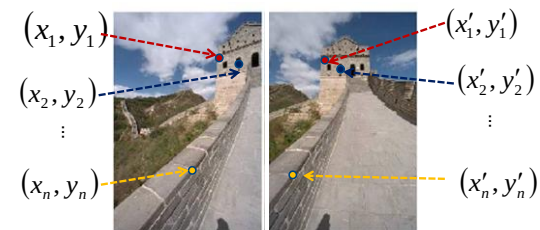
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Source: Aljoscha Ethos

86

Homography



To **compute** the homography given pairs of corresponding points in the images, we need to set up an equation where the parameters of **H** are the unknowns...

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Wolfram Grauert

87

Solving for homographies



$$\begin{bmatrix} wx'_i \\ wy'_i \\ w'_i \end{bmatrix} \cong \begin{bmatrix} h_{00} & h_{01} & h_{02} \\ h_{10} & h_{11} & h_{12} \\ h_{20} & h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

Solving for homographies



$$\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x'_1 x_1 & -x'_1 y_1 & -x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -y'_1 x_1 & -y'_1 y_1 & -y'_1 \\ & & & & & & & & \\ x_n & y_n & 1 & 0 & 0 & 0 & -x'_n x_n & -x'_n y_n & -x'_n \\ 0 & 0 & 0 & x_n & y_n & 1 & -y'_n x_n & -y'_n y_n & -y'_n \end{bmatrix} \begin{bmatrix} h_{00} \\ h_{01} \\ h_{02} \\ h_{10} \\ h_{11} \\ h_{12} \\ h_{20} \\ h_{21} \\ h_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{A} \quad \mathbf{h} \quad \mathbf{0}$$

$$2n \times 9 \quad 9 \quad 2n$$

Defines a least squares problem: minimize $\|\mathbf{A}\mathbf{h} - \mathbf{0}\|^2$

- Since **h** is only defined up to scale, solve for unit vector $\hat{\mathbf{h}}$
- Solution: $\hat{\mathbf{h}}$ = eigenvector of $\mathbf{A}^T \mathbf{A}$ with smallest eigenvalue

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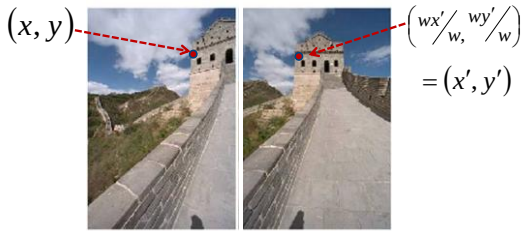
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Source: Aljoscha Ethos

89

Homography

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To **apply** a given homography H

- Compute $\mathbf{p}' = H\mathbf{p}$ (regular matrix multiply)
- Convert $\mathbf{p}'$ from homogeneous to image coordinates

$$\begin{bmatrix} wx' \\ wy' \\ w \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$\mathbf{p}' \quad H \quad \mathbf{p}$

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11

RANSAC for estimating homography

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- RANSAC loop:
 1. Select four feature pairs (at random)
 2. Compute homography H (exact)
 3. Compute *inliers* where $SSD(p_i', H p_i) < \epsilon$
 4. Keep largest set of inliers
 5. Re-compute least-squares H estimate on all of the inliers



Slide credit: Steve Seitz

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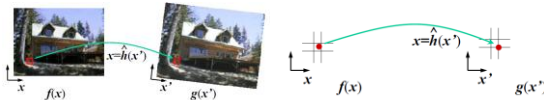
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12

Recall: Warping

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- GPU/OpenGL rendering
- Inverse map with interpolation:



procedure *inverseWarp*(f, h , out g):

For every pixel x' in $g(x')$

1. Compute the source location $x = \hat{h}(x')$
2. Resample $f(x)$ at location x and copy to $g(x')$

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13

Recap: How to stitch together a panorama (a.k.a. mosaic)?

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Basic Procedure

- Take a sequence of images from the same position
 - Rotate the camera about its optical center
- Compute transformation (homography) between second image and first using corresponding points.
- Transform the second image to overlap with the first.
- Blend the two together to create a mosaic.
- (If there are more images, repeat)

Source: Steve Seitz

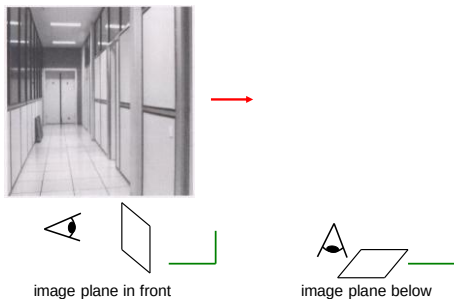
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14

Image warping with homographies

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Source: Steve Seitz

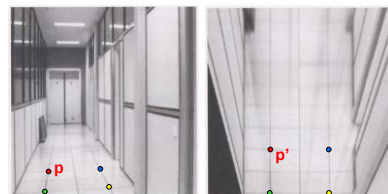
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15

Image rectification

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Source: Steve Seitz

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16

Image warping with homographies

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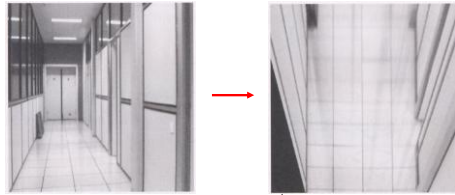


image plane in front

image plane below

black area
where no pixel
maps to

Source: Steve Seitz

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Image warping with homographies

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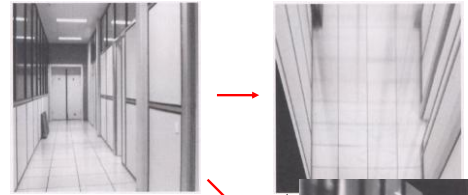


image plane in front

image plane below

black area
where no pixel
maps to

Source: Steve Seitz

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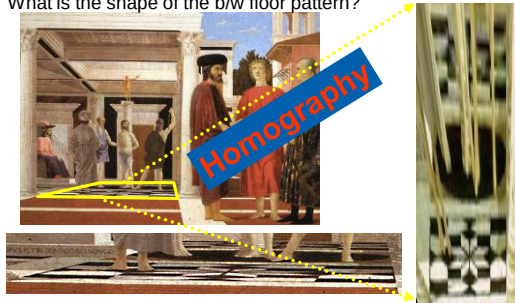
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Analysing patterns and shapes

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What is the shape of the b/w floor pattern?



The floor (enlarged)

Automatically
rectified floor

Slide from Antonio Criminisi

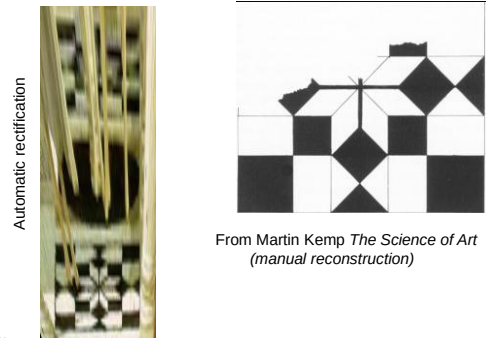
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Analysing patterns and shapes

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Automatic
rectificationFrom Martin Kemp *The Science of Art*
(manual reconstruction)

Slide from Antonio Criminisi

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Analysing patterns and shapes

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*St. Lucy Altarpiece*, D. Veneziano

Automatically rectified floor

Slide from Criminisi

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91

Analysing patterns and shapes

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Automatic
rectificationFrom Martin Kemp, *The Science of Art*
(manual reconstruction)

Slide from Criminisi

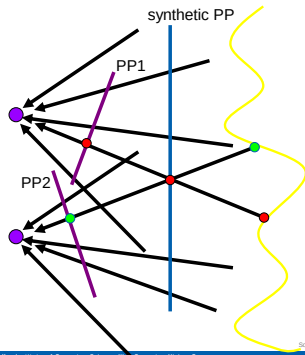
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92

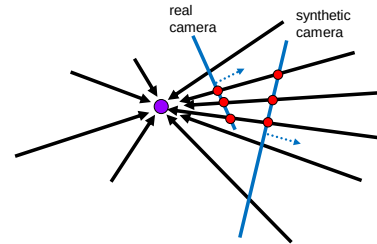
Changing camera center

Does it still work?



Source: Aljoscha Ethos

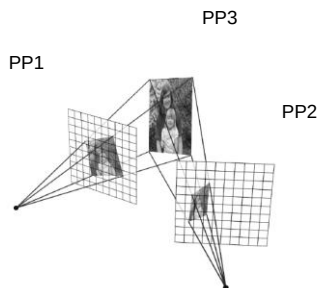
Recall: same camera center



Can generate synthetic camera view as long as it has the **same center of projection**.

Source: Aljoscha Ethos

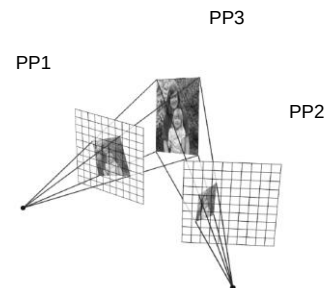
...or planar scene



- PP3 is a projection plane of both centers of projection, so we are OK!

Source: Aljoscha Ethos

...or planar scene (or far away)



- PP3 is a projection plane of both centers of projection, so we are OK!
- This is how big aerial photographs are made

Source: Aljoscha Ethos



Source: Aljoscha Ethos

Source: Aljoscha Ethos